



Wiruna Wanderings—April 2026

By Alessandro Spina

The April Wiruna weekend was a fantastic weekend. The weather forecast for the week leading up to New Moon was for clear skies. I took advantage of this and headed up on Thursday to try to sneak three nights of observing in. I set up my SkyWatcher dob alongside the Club's 17.5-inch dob. As the sun set conditions seemed ideal, mild temperature, no cloud, and little wind.

I decided that night would be a tour of Carina (the Keel of the mythical Argo Navis). First on the list was the 6th magnitude globular cluster **NGC 2808**. It is one of the Milky Way's most massive globulars (~1 million stars) and notably hosts five distinct stellar populations — a clue that some globulars are accreted dwarf-galaxy cores¹. To find it, use the Telrad to point the scope along the line between β -Carinae and ι -Carinae. As you pan the scope along that line, you'll pick the bright haze of the globular easily. Starting with a 16mm (94 $\times$ magnification, 0.87° TFOV), NGC 2808 is on the cusp of being fully resolved. With a 9mm (167 $\times$ magnification, 0.49° TFOV) the resolved stars come into focus (something that was much tougher to do in my 10-inch). Averted vision (AV) really helped to bring out more of the fainter stars in the halo. On one side, three chains of stars (perhaps foreground stars) extend out like “spiral arms” from the core. Next was **NGC 3114**, a 4th magnitude open cluster. To find this cluster, use the naked-eye visible IC 2602 (the “Southern Pleiades”) as a jumping off

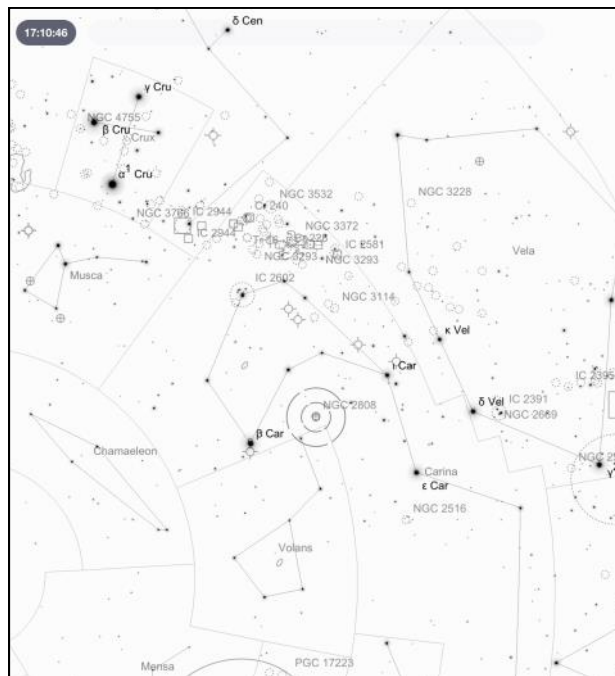


Figure 1: NGC 2808 GC in Car

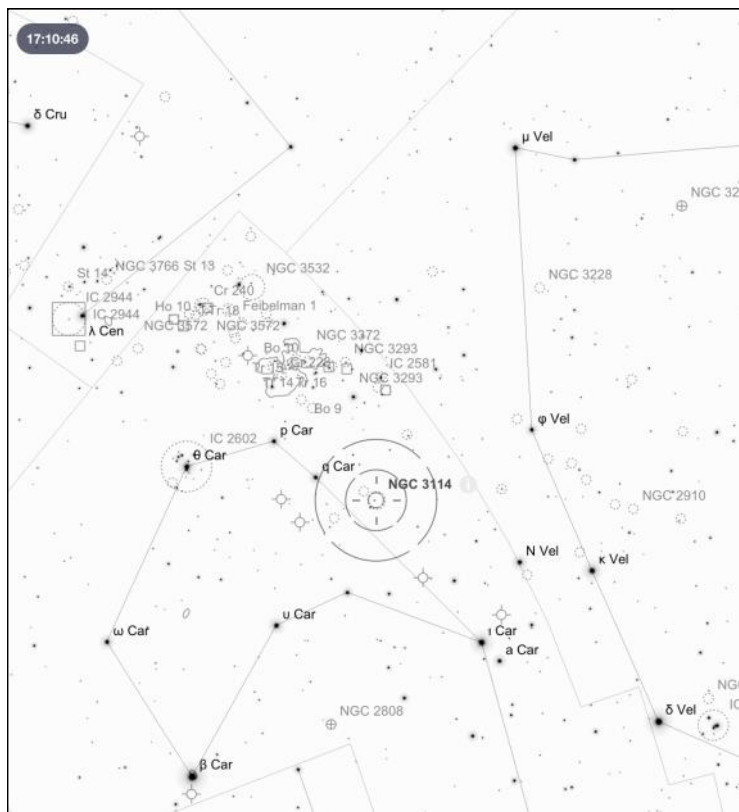


Figure 2: NGC 3114 OC in Car

point. Then draw a line through p-Carinae and q-Carinae, a further ~4° north sits NGC 3114. This cluster requires a large field of view (FOV), so I immediately switched to the 35mm (43 $\times$ magnification, 1.6° TFOV). **Southern Gems by Stephen O'Meara** (page 163) refers to the stars as forming the shape of a “radio telescope”. At first, I was sceptical as I struggled to pick out any shape in the eyepiece. The rich starfield makes it tricky to know where the cluster starts and ends. However, looking through the finderscope (with its ~5° FOV) makes the “radio telescope” shape abundantly clear. A triangle of bright stars formed the “base”, and semi-circular pattern of stars formed the “dish”. A bright yellowish star sits within the “dish”. I endorse the renaming of NGC 3114 as “The Radio Dish” cluster.

The next object, **NGC 3199** was also inspired by Southern Gems (page 170). It does not appear in my Cambridge Star Atlas; hence I had never noted this object before. This nebula is associated with a Wolf-Rayet star — WR 18, an exceptionally hot and luminous massive star whose extreme stellar winds have stripped away its outer hydrogen layers. NGC 3199, also known as the Banana Nebula, is a wind-blown bubble: a shell of stellar-wind gas sweeping up and ionizing the surrounding interstellar medium. Its impressive crescent shape was long attributed to a bow shock, though more recent work suggests the asymmetry reflects a denser pocket of the interstellar medium².

Swinging over to the other side of Vela, use the “False Cross” as your starting point to find **IC 2391** which is easy to pick up with the naked-eye just above δ -Velorum (see Figure 10). This cluster needs a low power eyepiece to capture all its members. I picked out 4 brighter stars in a Trapezium shape in the centre of the cluster, with trio of stars tucked inside with one member having a distinct yellow colour.

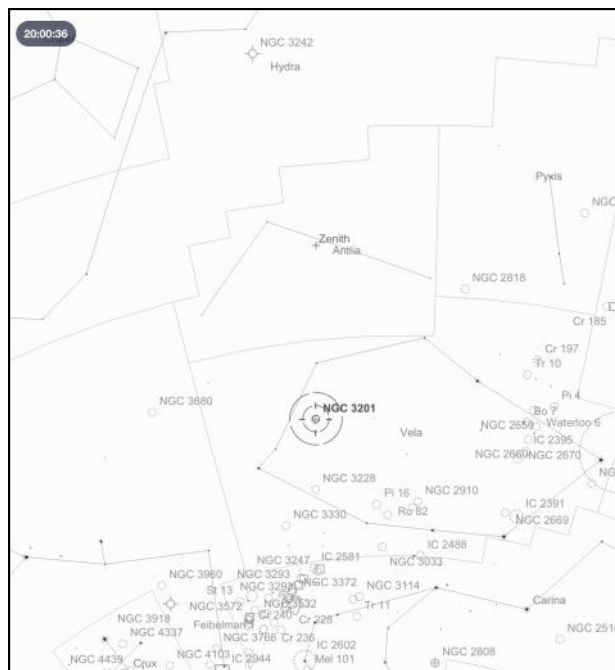


Figure 9: NGC 3201 GC in Vel

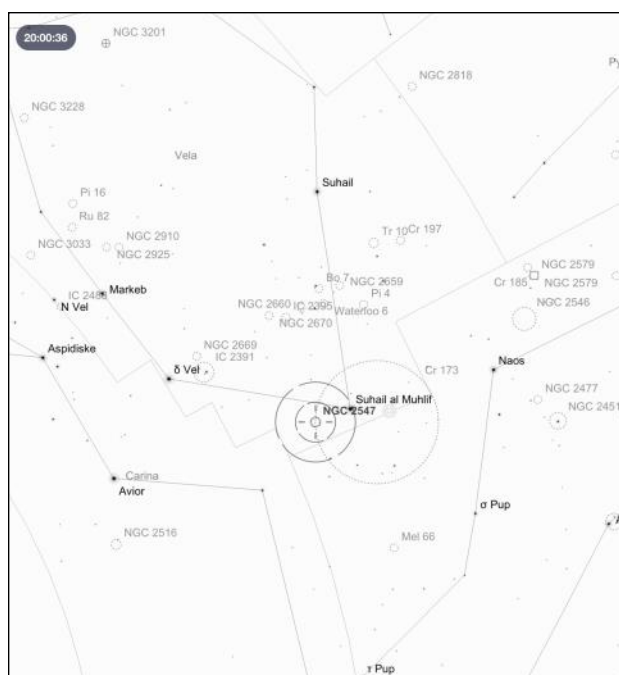


Figure 10: NGC 2547 OC in Vel

I then hopped over to Pyxis to hunt down another recommendation from Southern Gems (page 154), the planetary nebula **NGC 2818**. Pyxis itself is not a bright constellation, tucked in between Puppis and Vela. I could just make out the 'L-shape' combination of α , β , γ and δ - Pyxidis. Starting from γ -Velorum, cast your eyes north-west to ψ -Velorum, from there is a chain of 3 stars just visible to the naked eye which will guide you towards the planetary (see Figure 11). The intriguing thing about this planetary is it sits within an open cluster, like M46 in Puppis (though recent Gaia work suggests NGC 2818 may be physically associated with its host cluster⁹). The cluster itself is designated **Mel 96**.

Next was the open cluster **NGC 2547**. Start from the “False Cross” will give you χ -Velorum and δ -Velorum. Following this around to γ -Velorum (see Figure 10). γ -Velorum itself is interesting. It is a quadruple star system with its companion easily identified at 41 arc-seconds apart. The brighter component is itself a spectroscopic binary containing a blue supergiant and a Wolf-Rayet (WR) star — the closest WR star to Earth⁸. Two bright stars beneath point to the double star pair, completing the view. Using γ -Velorum as a marker in the finderscope, drift south $\sim 1^\circ$ and you’ll pick up a bright knot of stars which marks the cluster. NGC 2547 is a large cluster, 25' wide, but is relatively sparse. To my eye two overlapping oval chains of stars forming the bulk of the cluster. Continuing on, draw a line from γ -Velorum to λ -Velorum and halfway along this line you will find **NGC 2659** (see Figure 10). In the eyepiece this cluster formed a neat triangle with two other stars. All members of this cluster are fainter, giving it an overall lower surface brightness. But its irregular shape, and density are enough to pick it out from the field of view.

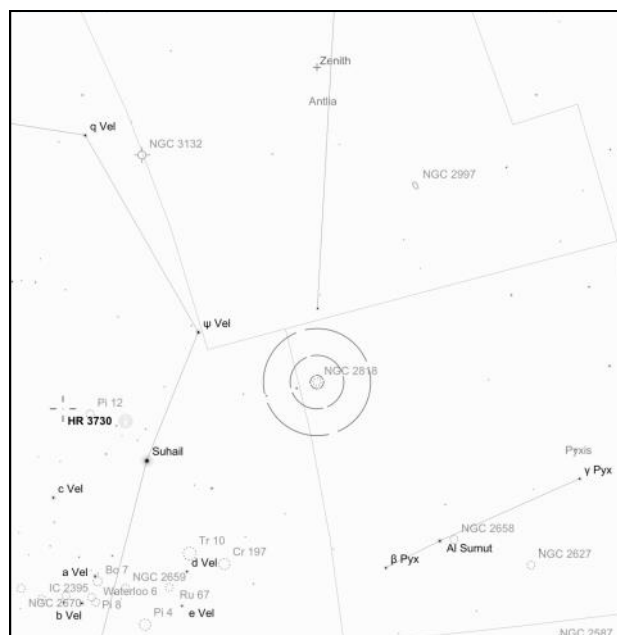


Figure 11: NGC 2818 PN + OC in Pyx

At approximately 8th magnitude, this is not a bright cluster, but it is rich enough to easily distinguish in the FOV. The cluster spans an area of 8'. In the middle of the cluster floats the grey, slightly elongated disk of NGC 2818. With averted vision, I get the sense the top-right/bottom-left taper off into a fainter extension of the disk. While in Pyxis I also decided to hunt down **NGC 2658**, made easy by its proximity to α -Pyxidis (see Figure 11).

By this time in the night Scorpius and Sagittarius were rising to the east. I spent the remainder of the night visiting some old favourites such as the Lagoon, Trifid Nebula and M55 with the benefit of the larger aperture.

Saturday brought additional members on site. Mark Notary set up the Club's solar scope (Figure 12). The Lunt solar scope uses a H α filter to isolate the hydrogen- α emission line, giving the sun's disk a deep orange-red colour. When I first looked through the solar scope at 10am several large prominences, caused by magnetic fields warping and twisting the sun's surface, could be seen dangling off the sun's disk to the bottom right. The top of the sun was also marked with several prominences that appeared to arc up and back onto the sun's surface. A collection of sunspots tracked above the sun's equator. As I wandered around doing things during day, I would occasionally check back on the solar scope and was amazed to see just how much movement there was as the prominences and sunspots tracked across the sun's disk. Well worth having a look through if you ever spot this solar scope out on the field.

Saturday evening brought ideal conditions. The wind had died down and there was no sign of any cloud. It would turn out to have the best seeing conditions of the three nights. So as sunset approached it was time to roll out the Club's new 22-inch Dob, Lord Sidious. Setting up this scope was straightforward. A quick two-star alignment was all that was needed to get the ArgoNavis and ServoCAT running. First object was M42 which was sinking towards



Figure 12: Lunt solar scope



Figure 13: The author measuring whether Lord Sidious will fit in his garage.

the western horizon. Satisfied with our alignment, we moved onto the first real test, the Horsehead Nebula. This was something I had never seen visually, so I was keen to test out Lord Sidious. I centred the telescope on NGC 2023, a reflection nebula just below the Horsehead (**Barnard 33**). Pushing Alnitak out of the FOV helped to reduce the glare from the bright star. With a 31mm eyepiece (90 $\times$ magnification, 0.91 $^\circ$ TFOV) and no filter in, I could immediately make out a dully illuminated nebula that stretched across the FOV with a distinctive curved silhouette sitting in the middle. Upping the magnification a little and blinking with the UHC filter helped bring out the shape even more. A chain of stars sitting above the nebula helped to increase our confidence that this was indeed B33. An exciting start to the night.

With a group of visitors and members (some old and new) hanging around the scope, we started on a Star Party style greatest-hits tour of the sky including, Eta Carinae, Tarantula Nebula, Omega Centauri, M65/M66 in Leo, M83, M4 and NGC 6231. One of the highlights of the night for me was Eta Carina. With a low-power eyepiece it always makes for a beautiful vista. Adding a UHC filter dialled up the contrast, making the view even more impressive. One of the many great features of Lord Sidious is its sliding rack that sits within the optical tube assembly which allows you to have 3 filters in place (we had a UHC, OIII and H-Beta) and simply push the slider up or down to change the filter. No unscrewing filters with cold hands and going up and down the ladder. The real magical views came with some higher magnification. Bumping up the power and focusing on Eta itself, revealed the bulbous Homunculus Nebula — the bipolar shell expelled during Eta Carinae's Great Eruption in 1843¹⁰. The shape and the dark lines that mark the expanding bubble were clear and well resolved. I was instantly reminded of one of my favourite

Wiruna memories from many years ago — sitting around Mike Kerr's then-new 25-inch Obsession, waiting for my turn at the eyepiece. With Pink Floyd playing in the distance, a younger me was called to the eyepiece for my first ever look at the Homunculus. 20 years later I was transported back to that moment.

Michael, a fellow visual observer visiting from Canberra, suggested **NGC 6520**, the "**Deadman's Cluster**" (See Figure 14). I was not familiar with this open cluster in Sagittarius, so we swung down to the eastern sky where Sagittarius had cleared the trees. We were not disappointed. The cluster itself is rich and compact, with a tight circle of stars surrounding an inner bright star. Beneath that sits a dark nebula Barnard 86 ("Ink Spot"). The field is also marked by a bright orange star next to B86. Together this makes for a beautiful view with contrasting colours and darkness. Michael also suggested a nearby pair of globulars, **NGC 6528** and **NGC 6522** which sits in patch of sky referred to as Baade's Window. As I learned Baade's Window is an area of the sky with relatively low amounts of interstellar dust. This area is considered an observational "window" to the Galactic Centre of our galaxy. With a low-powered eyepiece in, you can comfortably fit both globulars in the same FOV.

As the crowd thinned out, I seized the opportunity to confirm some objects from my previous night's observing. First up, I punched **NGC 3199** into the Argo Navis control panel to confirm if this was the object I had found on Thursday night. To my delight, it was indeed, although much brighter and with more intricate detail in the nebula itself. With the extra aperture, I could also see now that the arc did extend further down. I also punched in **NGC 3132**, the planetary in Vela that had eluded me the previous night. Given its brightness, it's shocking that I couldn't find this the night before. This is a beautiful planetary nebula with its soft bluish-grey tinge and 10th magnitude central star immediately apparent. The central star is itself a binary, with its companion providing the energy to illuminate the planetary. NGC 3132 is variously called the "Eight-Burst Nebula" or "Southern Ring Nebula", for its eight petal-like extensions and ring shape.

Sunday morning arrived and it was time to pack up and drive home unfortunately. With 3 nights of observing, through 3 different telescopes it was a marvellous weekend. Until next month. Clear skies¹¹.

Table of Targets

Name	Type	Constellation	RA (J2000)	DEC (J2000)	Mag	Size
NGC 2808	Globular Cluster	Carina	09h 12m 03s	-64° 51' 46"	5.7	9.0'
NGC 3114	Open Cluster	Carina	10h 02m 30s	-60° 07' 50"	4.2	12'
NGC 3199	Emission Nebula	Carina	10h 16m 33s	-57° 56' 02"	xx	20' × 15'
NGC 3293	Open Cluster	Carina	10h 35m 50s	-58° 14' 04"	4.7	5.1'
NGC 3532	Open Cluster	Carina	11h 05m 34s	-58° 42' 13"	3.0	55'
NGC 3572	Open Cluster	Carina	11h 10m 32s	-60° 17' 23"	6.6	4.1'
NGC 3590	Open Cluster	Carina	11h 12m 59s	-60° 47' 20"	8.2	5.1'
IC 2714	Open Cluster	Carina	11h 17m 27s	-62° 43' 30"	8.2	7.2'
Mel 105	Open Cluster	Carina	11h 19m 40s	-63° 29' 16"	8.5	6.5'
IC 2602	Open Cluster (Southern Pleiades)	Carina	10h 42m 57s	-64° 23' 39"	1.9	50'
Mel 101	Open Cluster	Carina	10h 42m 08s	-65° 06' 35"	8.0	16'
NGC 3766	Open Cluster	Centaurus	11h 36m 14s	-61° 36' 19"	5.3	12'
NGC 3201	Globular Cluster	Vela	10h 17m 37s	-46° 24' 40"	8.2	9.6'
NGC 2547	Open Cluster	Vela	08h 10m 02s	-49° 12' 08"	4.7	7.8'
IC 2391	Open Cluster (Omicron Vel)	Vela	08h 40m 32s	-53° 02' 08"	2.5	29'
NGC 2659	Open Cluster	Vela	08h 42m 33s	-45° 00' 02"	8.6	5.1'
NGC 3132	Planetary Nebula	Vela	10h 07m 02s	-40° 26' 12"	10.0	0.5'
NGC 2818	Planetary Nebula + OC	Pyxis	09h 16m 02s	-36° 37' 37"	11.6	0.8'
Horsehead Nebula	Dark Nebula (B33 / IC 434)	Orion	05h 40m 59s	-02° 27' 30"	7.3	6.0' × 4.0'
NGC 3372	Emission Nebula (Eta Carinae)	Carina	10h 45m 09s	-59° 52' 00"	1.0	120'
NGC 6520	Open Cluster (Dead Man's Chest)	Sagittarius	18h 03m 24s	-27° 53' 10"	7.6	5.4'
NGC 6528	Globular Cluster (Baade's Window)	Sagittarius	18h 04m 50s	-30° 03' 21"	10.7	3.6'
NGC 5286	Globular Cluster	Centaurus	13h 46m 27s	-51° 22' 27"	7.3	11'
NGC 6522	Globular Cluster	Sagittarius	18h 03m 34s	-30° 02' 02"	9.5	9.5'

¹ <https://phys.org/news/2022-12-astronomers-chemical-composition-multiple-stellar.html>

² <https://arxiv.org/abs/1708.02177>

³ <https://phys.org/news/2022-08-young-cluster-ngc.html>

⁴ I found this image on AstroBin which nicely captures the region: <https://app.astrobin.com/i/221c0b>

⁵ <https://academic.oup.com/mnras/article/467/4/3864/2982876?login=false>

⁶ <https://academic.oup.com/mnras/article/404/3/1203/1049211?login=false>

⁷ <https://arxiv.org/abs/2202.01599>

⁸ https://www.aanda.org/articles/aa/full_html/2014/03/aa23288-13/aa23288-13.html

⁹ https://www.aanda.org/articles/aa/full_html/2025/04/aa53031-24/aa53031-24.html

¹⁰ See 3D model here: <https://svs.gsfc.nasa.gov/11568/>

¹¹ Have any suggestions for interesting deep sky objects I should tackle? Drop me an email at: aspi624@gmail.com.